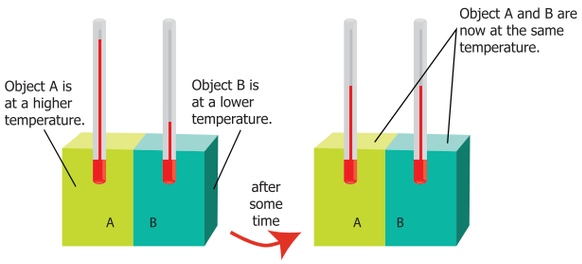
determines the direction in which **'heat'** will flow. Heat energy flows from higher temperature to lower temperature.

Thermal contact and Thermal equilibrium



▲Two objects at different temperatures are put together

Consider two bodies A and B. Let the temperature of A be higher than that of B. On bringing bodies A and B in contact , heat will flow from hot body A to the cold body B. Heat will continue to flow till both the bodies attain the same temperature.

The temperature determines the direction of flow of heat.

1. You are holding a hot cup of coffee. Would the Heat energy transfer from



   a. Your body to the coffee, or

   b. The coffee to your body?



2. You are standing outside on a summer day. It is 40°C outside (note that normal body temperature is 37°C). Would the Heat energy transfer from.

   a. Your body to the air particles, or

   b. The air particles to your body?



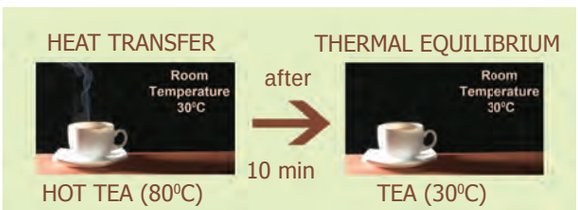
3. You are standing outside on a winter day. It is 23°C outside. Would the heat energy transfer from:

   a. Your body to the air particles, or

   b. The air particles to your body?

Two objects are said to be in **thermal contact** if they can exchange heat energy. **Thermal equilibrium** exists when two objects in thermal contact no longer affect each other's temperature.

For example, if a pot of milk from the refrigerator is set on the kitchen table, the two objects are in thermal contact. After certain period, their temperatures are the same, and they are said to be in thermal equilibrium.



1.7 Expansion in solids

Sam is trying to open a tight jar, but he cannot open it. He asks his uncle to help. His uncle says that pour some hot water on the lid of the jar. Sam does so and tries to open it now. Wow! The jar is opened easily!

Do you have such experience? How do you open a tightly closed cap of the